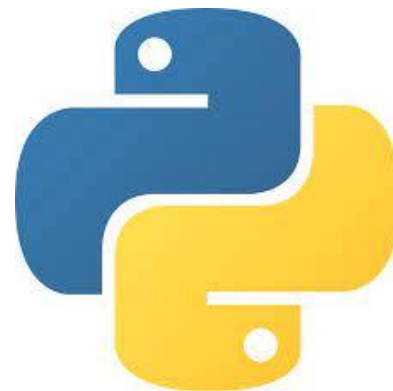
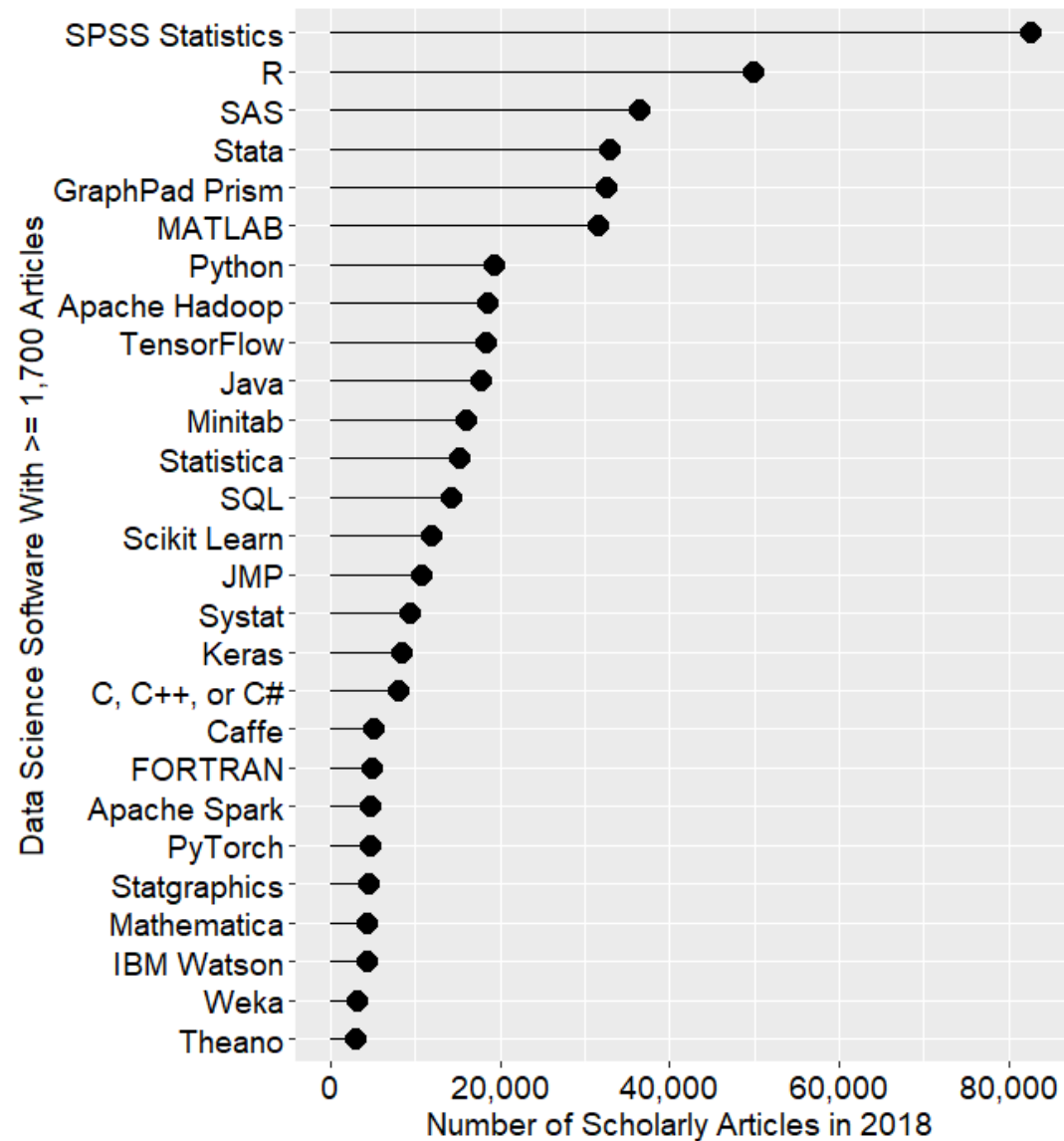
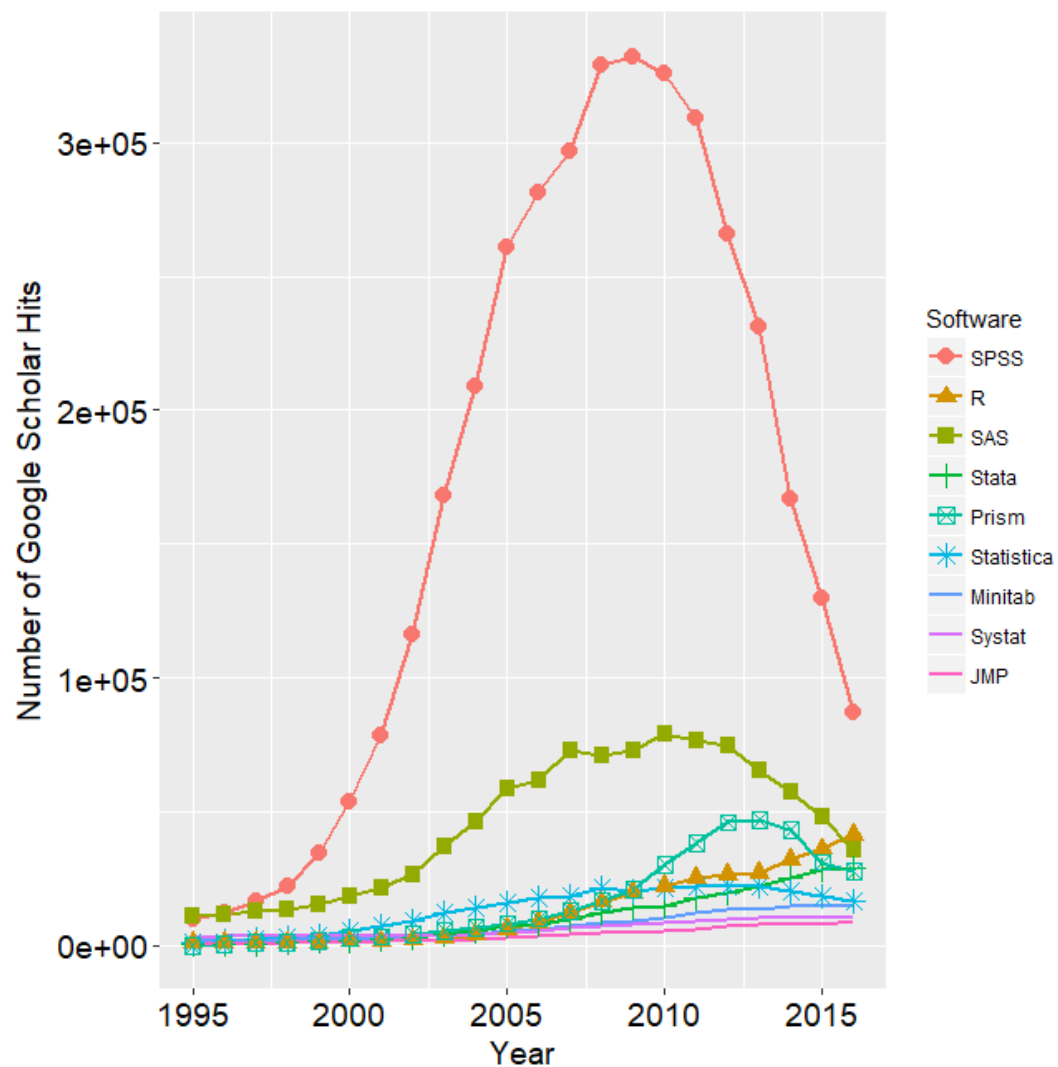




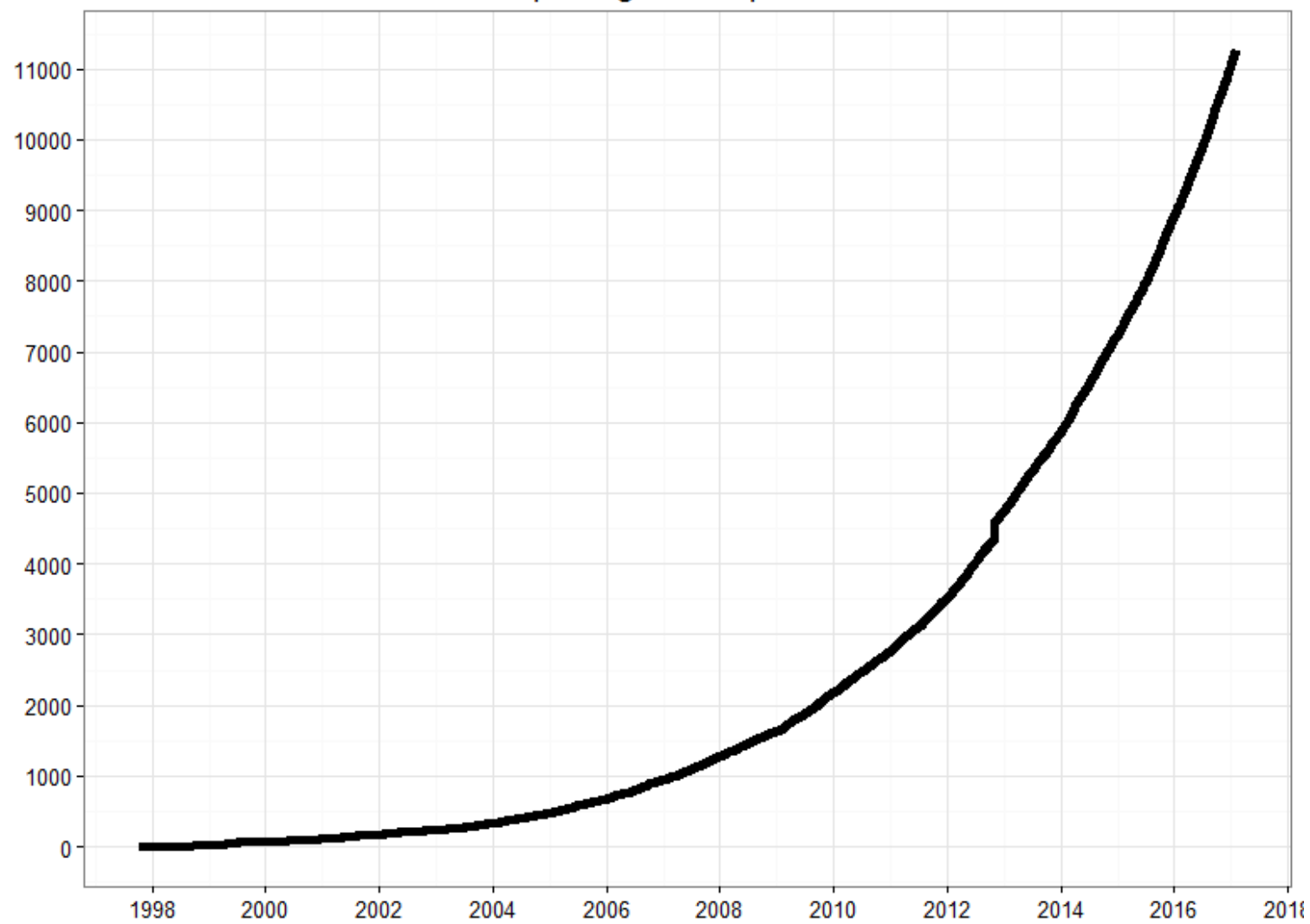
Tim Houle, PhD

Open-Source Statistical Software





Number of R packages ever published on CRAN



Software Guidance

- **United States Food and Drug Administration (FDA)**
 - General Principles of Software Validation; Final Guidance for Industry and FDA Staff (2002)
 - Guidance for Industry - Computerized Systems Used in Clinical Investigations (2007)
 - Statistical Software Clarifying Statement (2015)
- **International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals in Human Use (ICH)**
 - ICH E9 - Statistical Principles for Clinical Trials



Software Guidance

FDA

“FDA does not require use of any specific software for statistical analyses, and statistical software is not explicitly discussed in Title 21 of the Code of Federal Regulations [e.g., in 21CFR part 11]. However, the software package(s) used for statistical analyses should be fully documented in the submission, including version and build identification.”

E9 Statistical Principles for Clinical Trials

“The credibility of the numerical results of the analysis depends on the quality and validity of the methods and software (both internally and externally written) used both for data management (data entry, storage, verification, correction and retrieval) and also for processing the data statistically. Data management activities should therefore be based on thorough and effective standard operating procedures. The computer software used for data management and statistical analysis should be reliable, and documentation of appropriate software testing procedures should be available.”

R: Regulatory Compliance and Validation Issues
**A Guidance Document for the Use of R in Regulated Clinical
Trial Environments**

October 18, 2021

The R Foundation for Statistical Computing
c/o Institute for Statistics and Mathematics
Wirtschaftsuniversität Wien
Welthandelsplatz 1
1020 Vienna, Austria

Tel: (+43 1) 31336 4754

Fax: (+43 1) 31336 904754

Email: R-foundation-board@R-project.org

PhUSE US Connect 2019

Paper SA04

Submission of Software Programs to Regulatory Agencies

Maria Dalton, GlaxoSmithKline, Collegeville PA, USA

- <https://www.lexjansen.com/phuse-us/2019/sa/SA04.pdf>